The Full Generic Binary Tower of $X^2 + 1$: Complete Cycle Indices and Galois-Cover Genus

HCS-C393 theorem and reproducibility package

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Abstract

The polynomial X^2+1 generates the full automorphism group of its generic binary preimage tree at every height. A new critical value produces one sibling transposition; conjugation and restriction then force every last-level swap, without relying on a finite Galois computation. We derive the complete cycle-index recursion, the exact fixed-leaf law, and the genus of each smooth projective Galois-closure curve. The probability that a uniform height-n automorphism fixes a leaf is asymptotic to $2/n$, although its expected fixed-leaf count is exactly one. Exact small-tree permutations, critical-value identities and residue-field computations audit the normalization and its failure boundaries. These are source-local reconstructions with classical Galois and periodic-density owners. A transcendental generic basepoint, a Frobenius cycle and an original-map periodic orbit are not interchangeable, and no target prime-period or spectral identity is asserted.

Keywords: arboreal Galois group; quadratic dynamics; ramification; wreath product; cycle index; periodic density.

中文摘要

本文对二次映射的泛型逆像树证明每一层 Galois 群都是完整二叉树自同构群。 每层新增临界值给出一个兄弟交换，再由共轭产生全部末层交换。 进一步推导所有循环类型的精确递推、固定叶概率及 Galois 闭包曲线亏格； 固定叶概率渐近为二除以树高，而平均固定叶数始终为一。 文中区分泛型基点、数值特化、原映射时间和 Frobenius 时间， 经典理论明确归属，不把算术源结构写成目标零点对应。
关键词：树状Galois群；二次动力学；分歧；圈积；循环指标；周期点密度。

1 An intrinsic arithmetic source, not a fitted prime clock

Let $f(X) = X^2 + 1$, let t be transcendental over $\mathbb{Q}$, and let K_n be the splitting field of $f^n(X) - t$ over $\mathbb{Q}(t)$. The rooted tree T_n has root t, level-j vertices $f^j(X) = t$, and edges $x \mapsto f(x)$. Write

$$G_n = \mathrm{Aut}(T_n), \qquad d_n = |G_n| = 2^{2^n-1}.$$

The generic basepoint is part of the frozen source. No theorem below says that every specialization $t = a \in \mathbb{Q}$ has this Galois group.

Odoni [1] established the Galois theory of iterated polynomials. Pink [2] studies quadratic iterated monodromy with infinite postcritical orbits; not every quadratic rational map in that broader class has a full tree group. We give the specific new-branch proof needed for this polynomial. The wreath-product approach to periodic proportions belongs to Juul–Kurlberg–Madhu–Tucker [3]. Our finite audit and explicit source comparison do not certify global literature novelty.

Within the project, C374 treated the numerical basepoint-two power-map tower, with an affine index-two image. The present source has full nonabelian wreath growth, new ramification at every height and an original finite-field dynamical consequence. It is not another parameter table for that affine tower. All claims below concern native arithmetic; they do not insert a prime list or a logarithmic roof into the map.

2 The complete group from one new branch

Set $c_0 = 0$ and $c_i = f^i(0)$, so that

$$0, 1, 2, 5, 26, \ldots$$

are distinct: $c_{i+1} = c_i^2 + 1 > c_i$ for $i \geq 1$. The chain rule gives

$$(f^n)'(X) = 2^n \prod_{j=0}^{n-1} f^j(X). \tag{1}$$

Thus the finite branch values at height n are precisely $c_1, \ldots, c_n$. The rational function f^n has degree 2^n, so $[\overline{\mathbb{Q}}(X) : \overline{\mathbb{Q}}(f^n(X))] = 2^n$. Equivalently $f^n(X) - t$ is irreducible over $\overline{\mathbb{Q}}(t)$, and geometric monodromy is transitive on its leaves.

Theorem 1. *For every $n \geq 1$, both the geometric and arithmetic Galois groups of $f^n(X) - t$ are G_n. The restriction to height $n-1$ is onto with kernel $(C_2)^{2^{n-1}}$. The infinite tower has group $\mathrm{Aut}(T_\infty)$.*

Proof. At $t = c_n$ the only critical point in the fibre is $X = 0$. Indeed $f^j(X) = 0$ for $j \geq 1$ would imply $f^n(X) = c_{n-j} \neq c_n$. Near zero,

$$f^n(X) = c_n + A_n X^2 + O(X^4), \qquad A_n = 2^{n-1} \prod_{i=1}^{n-1} c_i \neq 0.$$

Local inertia is therefore a single transposition of two leaves. They are siblings because their common image under f is 1. The previous-height cover is unramified at this new branch value, so the transposition belongs to the last-level restriction kernel.

Inductively the group at height $n-1$ is G_{n-1}; restriction between splitting fields is surjective. The new kernel is contained in the group of independent swaps of the 2^{n-1} sibling pairs. Conjugate the one new swap by elements transitive on their parents. Every sibling swap is obtained, and these generate the entire kernel. The degree-two case starts the induction.

The arithmetic group contains the full geometric group and embeds in G_n, so it too is full. Thus the constant field is $\mathbb{Q}$. Compatible surjective restrictions give the full inverse limit: an automorphism of the union is exactly a family of compatible finite-level automorphisms. □

This proof derives the kernel rather than inferring it from a group-order fit. It also records why a post-critical collision would break the argument.

3 All branch inertia and the Galois-closure genus

At $t = c_i$, the ramified points in the degree-2^n covering are the 2^{n-i} roots of $f^{n-i}(X) = 0$. They are simple because no earlier critical value is zero. Each has ramification index two. The finite inertia permutation therefore has

2^{n-i} two-cycles, $\qquad 2^n - 2^{n-i+1}$ fixed leaves.

At infinity the polynomial has a single point of index 2^n, with inertia a 2^n-cycle. All inertia in the Galois closure is tame and there are no further branch values.

Theorem 2. *The smooth projective Galois-closure curve of $K_n/\mathbb{Q}(t)$ has genus*

$$g_n = 1 + \frac{d_n}{2}\left(\frac{n}{2} - 1 - 2^{-n}\right). \tag{2}$$

Proof. For a tame Galois cover of degree d_n, each base branch point of inertia order e contributes $d_n(1-1/e)$ to Riemann–Hurwitz. There are n finite branch values with $e = 2$ and one at infinity with $e = 2^n$. Consequently

$$2g_n - 2 = d_n[-2 + n/2 + (1 - 2^{-n})],$$

which is (2). □

For example $g_1 = g_2 = 0$ and $g_3 = 25$. This is not the genus of the original polynomial-cover source, which is the projective line.

4 The full cycle index and fixed-leaf law

Let $Z_n(x_1, x_2, \ldots)$ be the mean cycle monomial of a uniform element of G_n, with $Z_0 = x_1$. The wreath decomposition gives the complete recursion

$$Z_n(x) = \frac{1}{2}\left[Z_{n-1}(x)^2 + Z_{n-1}(x_2, x_4, \ldots)\right]. \tag{3}$$

Without the top swap, the two independent cycle partitions add. With the swap, two steps restrict to the product of the two independent subtree permutations, which is uniform; every resulting cycle length doubles. This proves every coefficient, not only a fixed-root marginal.

For an element with m_l cycles of length l,

$$\#\,\mathrm{Fix}(g^r) = \sum_{l|r} l m_l, \qquad \det(I - uU_g) = \prod_l (1-u^l)^{m_l}.$$

Here U_g is the native finite leaf permutation. Frobenius repetition changes that permutation; it is neither tree height nor the original map's iterate period. The finite determinant is not a target Fredholm operator.

Theorem 3. *If δ_n is the probability of at least one fixed leaf, then*

$$\delta_0 = 1, \qquad \delta_n = \delta_{n-1} - \tfrac{1}{2}\delta_{n-1}^2, \qquad n\delta_n \longrightarrow 2.$$

The mean fixed-leaf count is one at every height.

Proof. A top swap fixes no leaf. Without it, a fixed leaf exists unless both independent subtrees have none. This proves the recursion. The positive sequence decreases to zero, since its limit satisfies $L = L - L^2/2$. Also

$$\frac{1}{\delta_n} - \frac{1}{\delta_{n-1}} = \frac{1}{2(1-\delta_{n-1}/2)} \longrightarrow \frac{1}{2}.$$

Cesàro summation proves $n\delta_n \to 2$. The expected count equals one by transitivity and the orbit-counting lemma, or directly by (3). □

Thus a vanishing chance of any fixed leaf coexists with constant mean count. The distinction is intrinsic to the full group and cannot be read from one sample Frobenius element.

5 Conclusion

New critical inertia closes the entire generic binary Galois tower. Its cycle indices and Galois-cover genus follow at every height, not only on the finite verification grid. This is one paper, not three publications; its present record is Round one. The accompanying proof ledger, independent checks and source audit delimit the complete source theorem from the unachieved target bridge.

References

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